NOT MEASUREMENT SENSITIVE

MIL-S-62101C
11 September 1991
SUPERSEDING
MIL-S-62101B
3 October 1983

MILITARY SPECIFICATION

SEMITRAILERS, LOWBED: TILT DECK, 15 TO 30 TON, 4 WHEEL, MODIFIED COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lowbed, tilt deck, four-wheel semitrailers having rated payloads of 15 to 30 tons. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in procurement documents.

1.2 Classification. The semitrailer shall be one of the following sizes, as specified (see 6.2):

Size 15 - 15 ton
Size 20 - 20 ton
Size 22 - 22 ton
Size 25 - 25 ton
Size 30 - 30 ton.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

*	TT-W-572	- Wood Preservation, Water-Repellent.

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Markings, Blank.
	MIL-W-3912	- Wood Parts, Fabricated: For Transport Vehicle Bodies and Similar Applications.

STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

*	MIL-STD-209	- Slinging Eyes and Attachment for Lifting and Tiedown, Military Equipment.
	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
	MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Trailer (DoD adopted).
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of the Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.8, the vehicle, components, assemblies and accessories to be delivered under the contract shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification. All vehicle items shall be as represented in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297.

* 3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flashers or the two systems shall be independent of each other.

3.1.1.5 Wood treatment. Wood, except apitong, shall be treated in accordance with MIL-STD-1223.

3.1.1.6 Additional data plates. Additional data plates depicting the maximum towing speeds, permissible loads and permissible load distributions (see 3.3 through 3.4), shall be provided. The plates shall indicate the maximum towing speeds with and without payload, the maximum concentrated payload for each four foot segment of the entire length of the semitrailer, including the gooseneck, and the longitudinal location, length and width(s) of the maximum

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concentrated loads permissible as specified in 3.3. Appropriate loading instructions and cautions, as applicable to the manufacturer's design, shall be furnished on the plates. A plate shall be provided on each side of the vehicle in a conspicuous location. The plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.7 Lifting and typing down attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

* 3.1.1.8 Lubrication chart. When specified (see 6.2), a lubrication chart shall be provided. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

* 3.2.2 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

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* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

* 3.3 Performance. The fully equipped semitrailer shall evidence no part failure or permanent set or interference between parts when towed at speeds as great as: 25 miles per hour (mph), with rated payload, over unimproved roads and other reasonably hard, uneven terrain; 40 mph with rated payload over highways; and 55 mph without payload over highways.

3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to the coupled towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

3.3.2 Brake performance. The service brakes shall stop the semitrailer tractor combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.3.3 Tracking ability. The semitrailer shall conform to the tracking ability requirements of Federal Motor Carrier Safety Regulation 393.70(a).

3.3.4 Loading capability. The semitrailer shall be capable of being loaded from the rear with a mobile load equal to the required payload capacity without part failure, deformation, permanent set, or interference in any part of the semitrailer.

3.4 Payload and distribution. The rated payload capacity shall be not less than 30,000 pounds for size 15 semitrailers, 40,000 pounds for size 20 semitrailers, 44,000 pounds for size 22 semitrailers, 50,000 pounds for size 25 semitrailers, and 60,000 pounds for size 30 semitrailers.

3.4.1 Weight distribution. Except for size 30, load bearing members shall be rated for the loads imposed with the rated payload evenly distributed between the rear of the gooseneck and the center line of the rear bogie. For size 30, load bearing members shall be rated for the loads imposed with the rated payload evenly distributed for not less than 12 feet rearward from the rear of the gooseneck.

3.5 Dimensions and clearances. The semitrailer, without payload, uncoupled from the towing vehicle, resting on its landing legs, on level ground, and with the platform level, shall conform to the dimensions specified in table I.

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TABLE I. Dimensions and clearances.

Characteristic	Sizes 15, 20 and 22	Sizes 25 and 30
Overall length, feet (maximum)	36	38
Overall width, inches (maximum)	96	96
Platform height, from the ground to top surface of tilting platform, inches (maximum)	40	45
Tilting platform width, inches, plus or minus 1 inch	95	95
Platform tilted angle, degrees, without beavertail (see 3.5.1), (maximum)	15	15
Platform tilted angle, degrees, with beavertail (see 3.5.1), (maximum)	9	9
Beavertail tilted angle, degrees (see 3.5.2) (maximum)	18	18
Length from rear of gooseneck to centerline of rear bogie, feet (minimum)	11	11
Length of level platform from back of gooseneck to end of platform, exclusive of beavertail, feet (minimum)	19	21
Upper fifth wheel height, from the ground to the underside of the upper fifth wheel plate, inches, plus or minus 1 inch	48	53
Distance from vertical centerline of kingpin to front of semitrailer, inches, plus 2 minus 0 inches	18	18
Landing wheel clearance, radius from vertical centerline of kingpin to any point 6 inches or more below upper fifth wheel plate inches (minimum)	82	82
Forward swing radius, from vertical centerline of kingpin to most distant point on semitrailer nose, inches (maximum)	60	60

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TABLE I. Dimensions and clearances. (continued)

Characteristic	Sizes 15, 20 and 22	Sizes 25 and 30
Ground clearance, to bottom of platform, inches (minimum)	15	15
*Ground clearance, to bottom of axles and suspension, inches (minimum)	8	8

3.6 Construction and stresses. Main frame members shall be adequately braced with a sufficient number of crossmembers to insure stability under all specified operating conditions, including mobile loads being applied from the rear end of the semitrailer in the tilt and travel positions. The main frame members shall run the full length of the load space. The rear crossmember shall extend the full width of the semitrailer. When the semitrailer is statically loaded with load distributed as specified in 3.4.1, the maximum fiber stress in any frame member shall be not greater than 50 percent of the frame material yield strength. Under these conditions, the frame deflection shall be not greater than 2 inches. All parts of the frame shall be completely electric-arc welded.

3.6.1 Frame splices. Where frame members are spliced, joints shall be reinforced to fulfill the dynamic and static loading requirements of 3.3 through 3.4.1. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

3.6.2 Stress analysis. Shear and moment diagrams with maximum deflection calculations for the semitrailer loaded in accordance with 3.4 shall be performed. The strength of the floor, crossmembers, and outside frame members shall not be included in calculations to compute the main frame maximum fiber stress, but the weight of the floor, crossmembers, and outside frame members imposed on the main frame shall be included in the total load imposed on the main frame. Stress analysis shall include complete analysis of the gooseneck kickup area.

3.7 Platform. The tilting platform shall have one piece longitudinal members (without splices, butt-welds, or joints) running the full length of the platform and full width crossmembers to insure stability. The tilting platform shall be straight without protrusions or cutouts. If wheel wells are necessary, they shall be covered with safety tread steel plates. The wheel well steel plates shall be installed and secured on the same level as the remainder of the platform. Under all operating and articulating conditions, there shall be no interference between the tires and frame members, platform or wheel well covers. With the semitrailer on level ground, under all

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tilting conditions, there shall be no interference between the tires and frame members, platform, or wheel covers. When a stationary area is provided ahead of the tilting platform, the stationary area shall be floored the same as the tilting platform and shall be on the same plane so as to provide a level load space. The space between the stationary area and tilting platform shall be kept to a minimum. If a stationary area is furnished, it shall be not more than 3 feet long, measured from the rear of the gooseneck rearward.

* 3.7.1 Flooring. The platform shall be floored with apitong, white or red oak, or white ash of 2-1/4 inch uniform thickness as specified in MIL-W-3912. The tolerance on wood thickness shall be plus or minus 1/8 inch. The average weight of apitong used shall be not less than 50 pounds per cubic foot (pcf), with none of the individual pieces weighing less than 47 pcf, with no more than 15 percent moisture content at time of weighing. Flooring shall be laid lengthwise, as close together as practicable, attached with corrosion-resistant countersunk bolts and bridge style floor clips to eliminate drilling frame members. The top of countersunk bolt heads shall be not less than 1/8 inch and not more than 1/4 inch below the top of the wood surface. Exposed ends and edges of the flooring shall be protected with metal reinforcements. Wood surface shall be not less than 1/4 inch higher than adjoining metal members to prevent metal-to-metal contact of the load and the semitrailer. Platform crossmembers and bracing shall be adequate to sustain the rated payload with the load distributed as specified in 3.4.1. When platform width extensions are specified (see 3.7.5) the two center boards shall be secured to permit their easy removal by one man without special tools. The removable boards shall be capable of being used in the extension brackets specified.

* 3.7.2 Platform extension. Either a beavertail or when specified (see 6.2), a total level platform extension shall be furnished. The beavertail or level platform extension shall be floored the same as the platform. When no beavertail is furnished, and with the platform in tilted position and the rear edge of the platform on the ground, the angle of the platform with horizontal (ground) shall conform to table I.

3.7.2.1 Beavertail. The beavertail shall be a nominal 60 inches in length. The beavertail shall have steel decking support members on not more than 18-inch centers oriented 90 degrees to the direction of the wooden flooring. When in the tilted position, with the rear edge of the beavertail on the ground, the angle of the platform with horizontal (ground) and the angle of the beavertail platform with horizontal (ground) shall be as specified in table I.

* 3.7.3 Stake pockets. The semitrailer shall have not less than eight stake pockets, four on each side. The stake pockets shall be equally spaced from the front to the rear of the loading space, except when interference with critical frame members occurs. The stake pockets shall be

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4 inches square and shall be located inside the inner edge of the outer rail of the loading space. Stake pockets shall be furnished with drains to prevent entrapment of water.

* 3.7.4 Lashing D rings. The semitrailer shall have fifteen lashing D rings, located as follows: four on each side of the platform; two on each side of the gooseneck, inside the outer edge; and three across the rear. Lashing D rings and their mountings shall be the heavy duty type capable of withstanding a pull of 10,000 pounds in any direction without permanent deformation, and shall not protrude beyond the outer edge of the platform when not in use.

* 3.7.5 Platform width extension. When specified (see 6.2), swing type side extension brackets for planking shall be furnished to provide at least 1 foot of additional width to each side of the semitrailer. The number of brackets per side and design shall be the manufacturer's commercial standard, except in no case shall the bracket center to center spacing be more than 2 feet. Provisions shall be made for folding and locking the brackets against the main frame rails when the brackets are not being used. The brackets shall be so located, vertically, that when the removable planks specified in 3.7.1 are in place, the top of the planks shall be level with the remaining planks in the platform.

3.7.6 Platform tilt control mechanism. The platform tilt control mechanism shall permit the platform to be lowered and raised into tilt or travel position without external motivation, other than from the operator or mobile payload (vehicle). When placed in the tilted position the platform shall remain tilted without the use of mechanical locks or continued effort by the operator. The mechanism shall control the speed of the tilt action of the platform from zero to maximum velocity with rated payload and shall be a double-acting hydraulic retarding device with a factory preset automatic regulator to control the platform tilting speed.

3.7.7 Tilt lock. The semitrailer shall be equipped with a manual locking device which will secure the front of the platform in the horizontal hauling position after loading. The safety lock release shall be accomplished manually. An automatic locking device, in addition to the manual locking device, shall be furnished.

3.7.8 Ramps. Ramps shall be furnished to bridge the gap between the ground and the top of the rear edge of the platform when the platform is tilted. The ramps shall be attached at the rear of the platform and shall be of the swing type that fold onto the platform. The ramps shall be approximately 24 inches wide, located on each side of the rear of the platform and shall be sloped not more than 18 degrees. The ramps shall be of nominal 2-inch hardwood reinforced with steel and provided with support devices.

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3.8 Gooseneck. The gooseneck shall be of sufficient length to provide the minimum landing wheel clearance (turning clearance) specified and narrow enough in width and shaped to meet the maximum forward swing clearance radius specified. A fixed flat gooseneck shall be furnished. The entire area between the gooseneck main frame members shall be enclosed for use as a tool box. The underside shall be enclosed with a steel plate not less than 0.188 inch thick. Drain holes shall be provided. The top of the gooseneck shall be completely covered with not less than three hinged lids, arranged to provide access to all storage space. The deck lids shall be of safety tread steel plate with a minimum thickness of 0.188 inch, exclusive of the raised tread. Ring or retractable handles shall be provided to raise each deck lid. Suitable hoops shall be provided to permit padlocking the deck lids. Air lines and electrical cables and components within the storage area shall be protected from damage by stowed equipment, but shall be accessible for servicing.

3.8.1 Full width gooseneck. When specified (see 6.2), the gooseneck shall be the full width of the tilting platform. The top of the gooseneck outside the tool box area shall be completely covered with steel plate of the safety tread type with a minimum thickness of 0.188 inch exclusive of the raised tread; or when specified (see 6.2), floored between and flush with the top of the frame the same as specified for the main platform.

3.9 Axles and suspension. The axle and suspension system shall be of rated capacity at least equal to the load imposed at the ground when the semitrailer is loaded with specified payload. Axle and suspension system shall consist of either walking beam or spring suspension or combination walking beam and spring suspension and shall mount four dual wheels. Clearance shall prevent interference between wheels and any other portion of the vehicle under the conditions specified. Difference in height, for specified sets of dual wheels, shall be measured at the center of the tread on the outer tire. Sufficient articulation shall be provided to maintain approximately equal loading on each wheel under each of the following conditions:

(a) With not more than two diagonally opposite wheels elevated 6 inches.
(b) With forward pair of dual wheels elevated 6 inches above rear pair.
(c) With rear pair of dual wheels elevated 6 inches above forward pair.

3.9.1 Oil lubricated axles. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with seals. The hubcaps shall have a window. The hubcaps shall incorporate provisions for venting or equivalent method of withstanding internal pressure buildup without leakage. These shall be provisions for replenishing the oil supply.

* 3.10 Wheels, rim, tires and tubes. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size of tires furnished. Except when tube type tires are specified in procurement documents,

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multi-piece rims shall not be furnished. Tire, rim size and disc wheel size shall be the same for all wheels on each vehicle. When specified (see 6.2), for size 15, 20 and 22, disc type wheels shall be furnished.

* 3.10.1 Tires. Tires shall be of the steel belted radial type. Tires shall be tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle loaded with specified payload. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations. The number of tires furnished on each semitrailer shall be eight.

3.10.2 Inner tubes. When tube-type tires are specified in acquisition documents, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided in accordance with Tire and Rim Association recommendations.

* 3.10.3 Spare wheel and rim. When specified (see 6.2), a spare wheel or rim, shall be furnished. The spare wheel or rim shall be the same size as those furnished on the semitrailer.

* 3.10.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same manufacture, size, load range (ply rating), and tread design as the tires installed on the semitrailer.

* 3.10.5 Carrier for spare tire assembly. When specified (see 6.2), a carrier for the spare tire assembly shall be installed in a readily accessible location which does not interfere with the loadspace or ground clearance of the semitrailer. Threaded fasteners, when used to secure a spare in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3 .11 Brakes. The brakes shall be of the full air internal expanding type and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45, 393.46, and 393.47. Brake linings shall be of nonasbestos material. The brake system shall include automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains or cables, air reservoirs (with remote operating control for the drain valve positioned above the bottom of the frame rail), brake chambers, relay emergency valve, and all other parts and connections required for a complete air brake system. Gladhands shall conform to SAE J318. The location of the gladhands shall be as recommended

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in SAE J702. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the brake system shall extend below the bottom of wheel rims to insure, in case of a flat tire, that the weight of the semitrailer will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

* 3.11.1 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. With no additional application, the parking brakes shall hold the semitrailer, with rated payload, on a 10 percent incline with the semitrailer headed first up and then down the incline. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.12 Upper fifth wheel. The upper fifth wheel plate shall completely cover a 36 inch diameter lower fifth wheel when the lower fifth wheel is coupled to the kingpin of the upper fifth wheel and the semitrailer is turned at all angles. The upper fifth wheel shall conform to Federal Motor Carrier Safety Regulation 393.70(b).

* 3.12.1 Upper fifth wheel plate. The upper fifth wheel plate shall be not less than 0.250 inch thick for semitrailers up to and including size 20 and not less than 0.375 inch thick for all larger sizes. The plate shall be reinforced on the upper surface through the use of commercial steel angles, channels, and ribs. The plate shall include an access hole to permit servicing the brake connections and electrical connectors when these components are not serviceable from the top.

* 3.12.2 Pickup plate. A turned up lip pickup plate, compatible with the size (payload) of the semitrailer, shall be provided as an extension of the upper fifth wheel plate. The pickup plate shall have a turned up lip not less than 2 inches high. The pickup plate shall extend sufficiently forward to protect the semitrailer from damage during coupling and uncoupling operations to and from a truck tractor. The pickup plate shall be integral with the upper fifth wheel plate.

* 3.12.3 Kingpin. A 2-inch diameter kingpin conforming to SAE J700 shall be furnished. The kingpin shall be fabricated of heat treated alloy steel.

* 3.13 Landing gear. Unless landing supports are specified (see 3.14), elevating landing gear shall be furnished. The landing gear shall consist of two legs of the vertically retractable, two-speed type. Both legs shall be operated and controlled by means of a single hand crank.

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Supports for the crank extension shaft and clips for holding the crank when folded shall be provided. Each leg shall be equipped with a steel landing pad. Each pad shall provide an area not less than 100 square inches. Landing legs shall be located on each side at the rear of the gooseneck. Each leg shal1 be capable of supporting its portion of the fully loaded semitrailer and shall be adequately braced to the lowest extremity of the fixed portion of each leg. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its function or mechanical efficiency. The maximum cranking effort to raise the fully loaded semitrailer shall be not greater than 100 pounds.

* 3.13.1 Range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate from 47 inches to not less than 53 inches from the ground for size 15, 20 and 22 trailers, and from 50 inches to not less than 56 inches from the ground for size 25 and 30 trailers. Minimum ground clearance with landing gear retracted shall be 10 inches.

* 3.13.2 Air powered landing gear. When specified (see 6.2), in addition to the requirements of 3.13 and 3.13.1 (less hand crank, etc.), the landing gear shall be air powered. The air powered landing gear system shall include a control box equipped with a 12-volt direct current (dc) motor driven pump, master control valve, leveling valve, booster, main oil tank, air side oil tank, key switch and mounting hardware.

* 3.14 Landing supports. When specified (see 6.2), landing supports shall be furnished in lieu of landing gear. Landing supports shall consist of two legs of the gravity telescopic type, capable of being fixed in various positions. Each leg shall be equipped with a steel landing pad. Each pad shall provide an area not less than 100 square inches.

3.14.1 Range of adjustment for landing supports. Landing supports shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate from 47 inches to not less than 53 inches from the ground for size 15, 20 and 22 semitrailers, and from 50 inches to not less than 56 inches from the ground for size 25 and 30 semitrailers. Landing support adjustment shall be in not more than 2-inch increments. Minimum ground clearance with landing supports retracted shall be 14 inches.

* 3.15 Lubrication systems. All moving parts normally requiring lubrication shall be provided with accessible pressure grease fittings. Pressure type lubrication fittings shall be readily accessible for servicing and shall conform to SAE J534. Pressure relief fittings shall be used where high pressure lubricating equipment could cause damage to seals.

* 3.16 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail (unless recessed and fully protected), or on the semitrailer

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bumper. All electrical wiring shall conform to SAE J1292. Clearance and identification lights shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools.

3.16.1 Receptacle, 12 volt direct current (dc). The front of the semitrailer shall be equipped with a 7 contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.16.2 Twelve-volt dc system. A 12-volt dc lighting system conforming to 3.16 and 3.16.1 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.16.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system independent of the 12 volt dc lighting system shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.16. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.16.2, except the 24-volt dc stop and turn lamps shall be combined.

3.16.4 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt dc system (see 3.16.2) from a towing vehicle equipped with a nominal 24 volt dc electrical system shall be provided.

3.16.5 Receptacle, 24-volt dc. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow).
Contact D. Connect to ground (white).
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The rest of the contacts shall not be connected. Circuits B and J on tactical trucks are combination stop, and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt 12-contact receptacle.

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3.16.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.16.4), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the 28 volt dc regulated voltage of the tactical truck to within the maximum rated voltage of the semi trailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any damage to resistors, wiring, or adjacent components.

* 3.17 Mud flaps. Mud flaps shall be provided to the rear of the rear wheels, in accordance with SAE J682. A metal strip not less than 0.125 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.250 inch wide by 0.094 inch thick shall be furnished at each bolt.

3.18 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.19 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each semitrailer for operational use including at least the following: adjustment of electrical and brake systems; inflation of all tires; and complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.20 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the best commercial practice.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4. QUALITY ASSURANCE PROVISIONS

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the kingpin and rear axles. The imposed loading on the kingpin and rear axles will be computed using curb weight and payload distributed as specified in 3.4.1. The calculated imposed load on the rear axles will be used to ascertain that the suspension, axles and tires are of adequate capacity to meet contract requirements.

4.3.2 Road test. The first production vehicle shall be road tested for a distance of not less than 20 miles at speeds specified in 3.3 and loaded as specified in 3.4 to determine conformance to the performance requirements of this specification. In addition, the semitrailer shall be loaded with one and one half times the specified payload weight evenly distributed in the loadspace and shall be examined to determine conformance to the static requirements of this specification. Upon completion of the road test, the landing gear shall be tested.

4.3.2.1 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle

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to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.3 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished (see 3.16.4), the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test the resistor assembly, including resistors, wiring and adjacent components, shall be examined for evidence of overheating, deterioration or damage.

4.3.4 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

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4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.7 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The vehicles covered by this specification are intended for non-tactical use in transporting track and wheel mounted construction equipment, tracked vehicles, machinery and other cargo.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of vehicle required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of the appropriate service for painting and marking (see 3.1.1.1).
(e) Identification of the appropriate service for marking (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Lifting and tying down attachments, if required (see 3.1.1.7).
(h) Platform extension (beavertail), if required (see 3.7.2).
(i) Platform width extension, if required (see 3.7.5).
(j) Full width gooseneck, if required (see 3.8.1).
(k) Gooseneck wood flooring, if required (see 3.8.1).
(l) Disc type wheels, if required (see 3.10).
(m) Spare wheel and rim, if required (see 3.10.3).
(n) Spare tire assembly, if required (see 3.10.4).
(o) Carrier for spare tire assembly, if required (see 3.10.5).
(p) Air powered landing gear, if required (see 3.13.2).
(q) Landing supports, if required in lieu of landing gear (see 3.14).
(r) Independent 24 volt dc lighting system, if required (see 3.16.3).
(s) Interconnected 24 volt dc lighting system, if required (see 3.16.4).

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6.3 Subject term (key word) listing.

Air suspension
Fifth wheel
Gooseneck
Landing gear
Welding.

6.4 Changes from previous issues. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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